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DEPARTMENT OF INFORMATION COMPUTING TECHNOLOGY

PROJECT REPORT

REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR DIPLOMA IN WEB DEVELOPMENT

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The success and final outcome of my project required a lot of guidance and assistance from many people, and I am extremely privileged to have got this along the completion of my project. Everything that I have done is only due to such supervision and assistance of my supervisor and I would not forget to thank her. I respect and thanks to my supervisor for providing me an opportunity to do the project under her supervision.

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Chapter 1: Introduction

1.1 Overview

The Smart Anti-Theft Door Access System is a security project using a microcontroller that enhances physical access control using RFID authenticating for authorized users and the system automatically captures a photo when a potential unauthorized user attempts access using an ESP32-CAM module. The photo is stored in a local SD card for record-keeping or documenting evidence. This project is intended to share an affordable offline and smart anti-intrusion solution for homes, small offices, or dormitory spaces.

1.2 Problem Statement

Conventional door locks are susceptible to unauthorized access such as key duplication or lock-picking. Most entry level RFID systems, meanwhile, do not provide any logs and therefore no evidence of potential intrusions. Without onboarding surveillance, there is no way to determine who tried to enter, or when, leaving customers with a major security gap if they are looking for an inexpensive and uncomplicated access control solution with a visual element.

1.3 Project Objectives

- To create a smart door lock system with RFID technology for user authentication.
- To integrate an ESP32-CAM to capture an image of an intruder and store it when the entry logging detects unauthorized access.
- To store these captured images locally on an SD card, so those images can be accessed and reviewed.
- To build a simple, reliable low-cost hardware prototype that can be deployed in the real world.
- To build a system that does not rely upon the cloud or the internet.

1.4 Project Scope

- Hardware: Arduino Uno, ESP32-CAM, RFID module (RC522), MicroSD card, electromagnetic door lock, buzzer
- Software: Arduino IDE programming for access control and image capture trigger
- Security response: System will detect unauthorized RFID tag access and respond with sound alert and image capture
- Output: Images stored as .jpg files on SD card labeled with timestamp or ID number

1.5 Project Justification

- Uses open-source hardware which is affordable and fairly simple to set up for students or small business
- Doesn't require Internet or third-party services, making it privacy-conscious and reliable offline
- Great example of integration between IoT security components with embedded programming
- Good as a realistic smart entry prototype and follows current trends with smart home and IoT

1.6 Project Constraint

1. Time frame: The project is to be completed before September 2025; Chapter 1 of the system must be implemented by the deadline.
2. Resources: For the project completion, script, programming skill and basic development concept are needed. The script of the website will reference a template on Internet. For the programming skill, this project is just required to know basic of CPP for development. As a developer, we should have the basic concept for development to reduce some trouble during development.
3. Knowledge: Lack of knowledge regarding Arduino may affect the progress of the project.

1.7 Project Timeline

A Gantt chart is a tried-and-true method of visualizing the steps, assignments, and length of any project. As a result, when project owners or managers start a new project, the first thing they'll do is generate a Gantt chart. It's a good way to keep track of everything that's going on. In addition, project timelines provide a quick, visual representation of how long the project will take.

Weeks / Milestones	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Planning														
Hardware purchasing & wiring (RFID + ESP32-CAM)														
Coding RFID access control and trigger logic														

Coding ESP32-CAM image capture & SD storage														
Integration testing with full system														
Prepare demo, test multiple scenarios														
Prepare documentation, results, and analysis														
Testing Output & Final Debug														
Report Writing														

Table 1: Gantt Chart

Following the building of the website, the remaining time will be devoted entirely to the documentation and preparation of the website's presentation.

Chapter 2: Literature Review

2.1 Introduction

This section of the research will provide the information which have been gathered. This information will be summarized and organized according to importance. All of the information will be extracted from reliable sources, mostly from online journals, articles and books.

The information gathered in this section will be used as reference when declaring the requirements and designing and implementing the management system.

This chapter will list the offline security systems, research studies on electronic locking mechanisms, RFID authentication, ESP32-CAM integration, and best practices in access control system design. The study's findings are summarized in the following below.

2.2 History of RFID Anti-Theft Door Access System

The advancement of anti-theft door access systems has been ongoing as a result of technology innovation and emergent security needs. Historically, security relied on mechanical lock and key mechanisms—simple but effective—but had limitations in key duplication and no access audit functionality, which led to people creating alternatives to identify valid entry. The transition to electronic solutions began in the mid-20th century with systems like punch cards and magnetic stripe readers. These early systems allowed for programmable access and basic logging functions, which introduced a meaningful shift in favor of controlled and auditable entry systems. [1]

The decade of the 1980s saw the introduction of proximity cards based on RIFD technology. These cards utilized a non-contact approach to authentication and provided much better durability and usability at the point of use than previous systems methods. This was a turning point for access control moving towards the broader adoption of smart card technologies. The continued evolution of access control saw in the 1990s, and early 2000s smart cards with microprocessors embedded within the card and biometric systems (i.e., fingerprint recognition and facial recognition) were offered. These systems met the demand for an increase in security in relation to user authentication. [2]

The literature also illustrates RFID's even earlier origins: it emerged in the mid-20th century for military purposes like “Identification Friend or Foe” transponders, and later developed into commercial applications in the 1970s. By the 1990s and early 2000s RFID had standardized, was widely used, and had gained acceptance in security, logistics, and identification applications.[3]

2.3 The advantages of RFID Anti-Theft Door Access System

An RFID-based anti-theft door access system offers several compelling benefits compared to traditional mechanical locks or less robust electronic methods:

1. Each RFID tag or card carries a unique, encoded identifier, making replication or unauthorized duplication extremely challenging. This significantly reduces the risk of false credentials being used to gain unauthorized access. [4]
2. RFID access operates contactless—users simply tap or wave a card or fob near the reader, enabling quick and seamless entry. This efficient mechanism is ideal for scenarios requiring rapid access, such as daily building use or high-traffic areas.[5]
3. RFID cards and tags are designed for long-term use and resist wear and tear far better than traditional keys or magnetic stripe cards. The readers themselves are often robust and weather-resistant, minimizing servicing requirements.[4]
4. RFID systems can easily grow alongside your security needs. Adding new users or updating access rights requires only reprogramming, without physical hardware changes. This includes revoking licenses, managing time-based access constraints, and adapting to organizational changes.[4]

2.4 Features of RFID Anti-Theft Door Access System

There has been significant advancement in door access control technology, especially with the introduction of RFID-based anti-theft systems. These systems combine security, convenience, and automation to ensure safe and efficient entry management. A good RFID Anti-Theft Door Access System should provide reliable protection, ease of use, and adaptability to different environments. According to a web article, the key characteristics of a good RFID Anti-Theft Door Access System include:

1. Secure authentication to prevent unauthorized access.
2. Fast and contactless entry for user convenience.
3. Tamper detection to alert administrators of forced entry attempts.
4. Access logs for tracking and monitoring entry history.
5. Integration capability with alarms, cameras, or other security devices.

2.5 Comparison of RFID Anti-Theft Door Access System with Other Door Lock Types

In recent years, door access technologies have evolved significantly, providing users with multiple options for securing their properties. While traditional locks still exist, modern solutions such as RFID door locks, keypad locks, biometric systems, and mechanical key locks offer varying levels of security, convenience, and cost-effectiveness. According to various security technology reviews, the key comparison points include ease of use, security level, maintenance requirements, and adaptability to different environments.

Mechanical Key Locks

- Advantages:
    - Low cost and widely available.
    - Easy to install and replace.
    - No need for electricity or batteries.
- Disadvantages:
    - Keys can be lost, stolen, or duplicated without permission.
    - Vulnerable to lock picking and bumping.
    - No access tracking capability.
    - Requires manual key management for multiple users.

Keypad Locks

- Advantages:
    - Eliminates the need for physical keys.
    - Easy to change passcodes when necessary.
    - Can be battery-powered or connected to mains.
- Disadvantages:
    - Passcodes can be guessed, shared, or observed (shoulder surfing).
    - Requires frequent code changes for security.
    - Battery replacement or power supply maintenance needed.
    - Limited access tracking unless connected to a smart system.

Biometric Locks (Fingerprint/Facial Recognition)

- Advantages:
    - High security using unique biological identifiers.
    - No keys or cards required.
    - Difficult to duplicate or forge biometric data.
    - Can store multiple user profiles.
    -

- Disadvantages:
    - Higher cost compared to other lock types.
    - May fail with dirty, wet, or injured fingers, or in poor lighting for facial recognition.
    - Requires regular cleaning and calibration.
    - Some users may have privacy concerns over biometric data storage.

RFID Anti-Theft Door Access System

- Advantages:
    - Contactless and quick access with RFID card/tag.
    - Cards/tags are difficult to duplicate compared to keys.
    - Can integrate with alarms, CCTV, and access logs for enhanced security.
    - Scalable for large facilities — multiple users can be added or removed easily.
    - Allows tracking of entry/exit times for security management.
    - Resistant to physical picking or tampering.
    - Can function with backup battery during power outages.
- Disadvantages:
    - Cards or tags can still be lost or stolen.
    - Dependent on electronic components, which may fail over time.
    - Initial installation cost is higher than mechanical locks.
    - Requires regular maintenance for optimal performance.

2.6 Techniques and Technologies

There are several technologies available for smart door access control systems. Traditional access control systems often use mechanical locks or magnetic stripe cards as forms of electronic access control, which can be less secure and more difficult to manage. Furthermore, advancements in RFID technology, microcontrollers, and network-based modules have led to increased security and efficiency in access control systems. The system we are using includes several components: the UNO microcontroller to serve as the brain of the system, the RC522 RFID reader to read user identification and confirm their access, the ESP32-CAM to take images of users trying to access the system, and a 12V electromagnetic lock for locking/unlocking the door. The system components are also supported by modules, such as the relay, buzzer, and LM2596 buck converter, which allow the distribution of power to all of the components, deliver alerting functions, and allow proper operation and functionality of the system. Consequently, the system described provides a reliable, efficient, and secure smart

door access system, that is not dependent upon traditional keys. The user experience of the smart door access system also allows for improved monitoring, archiving, and control of access into the facility.

Chapter 3: Methodology

3.1 Introduction

This chapter will explain and define software development life cycle (SDLC), and what each step means. The methodology used and how this methodology helps this project, and the approach towards building the project. This chapter will also explain the hardware and software requirement needed for the project to run.

3.2 Spiral Methodology (Model)

This project is using the Spiral Model as the development model. The Spiral Model takes the best of both iterative development and risk elements, so it is a very good model for projects where hardware and software are integrated, like the RFID Smart Anti-Theft Door Access System. The Architectural iteration process in the Spiral Model breaks down the system to small systematic modules that could be developed, tested and reviewed or refitted in every cycle before being jointly implemented.

The model has four major activities in every cycle:

- Identify objectives and requirements – At this stage, the functional requirements of each module are defined, such as RFID authentication, relay control for the electromagnetic lock, ESP32-CAM image capture, and system alert functions.
- Evaluate alternatives and analyse risks – Alternative approaches for implementing each function are studied. For example, choosing between using UNO or directly programming ESP32-CAM for RFID communication, and analysing risks like power supply instability, signal interference, or failed module communication.
- Development and verification – Each hardware and software component is developed and tested individually. For instance, first testing RFID tag detection, then verifying relay switching to control the lock, and finally integrating the ESP32-CAM to capture images during unauthorized access attempts.

- Planning the next cycle – After each successful test, the outcomes are reviewed, and improvements or changes are planned for the next cycle. This allows early detection of issues, minimizing risks and ensuring the system works as intended.

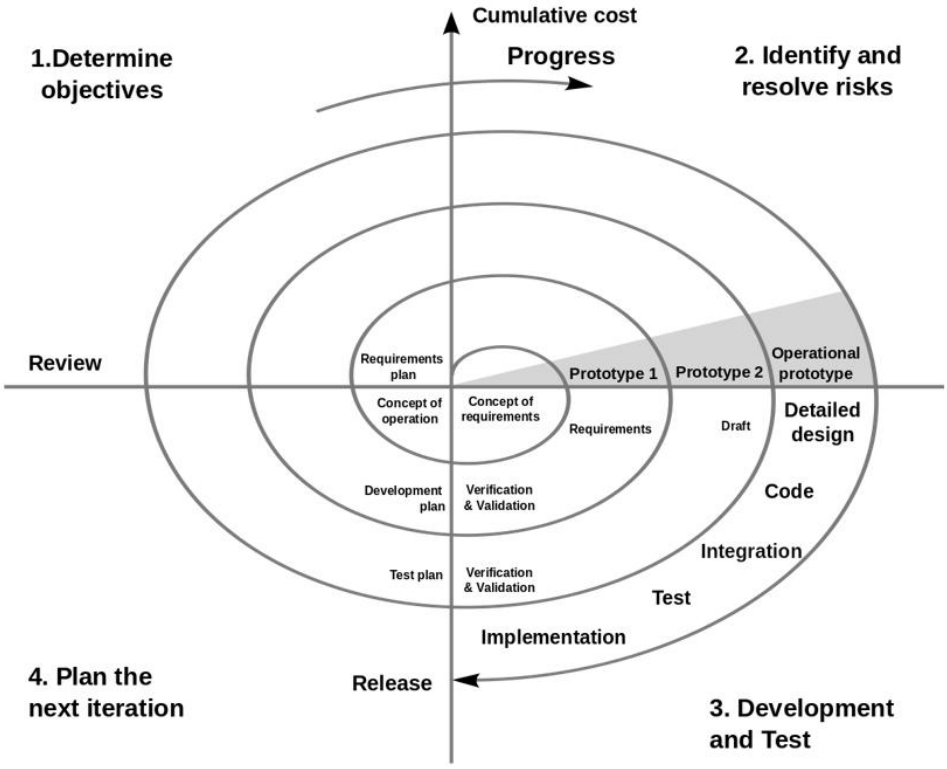
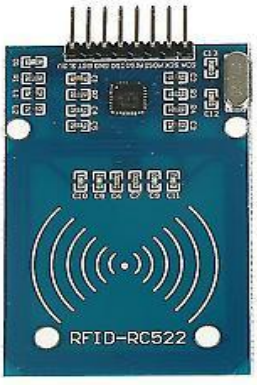


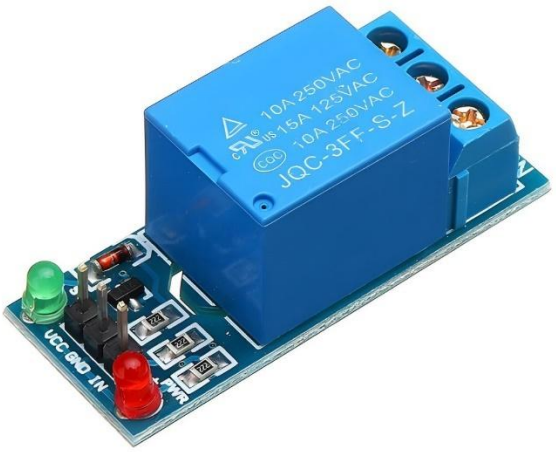
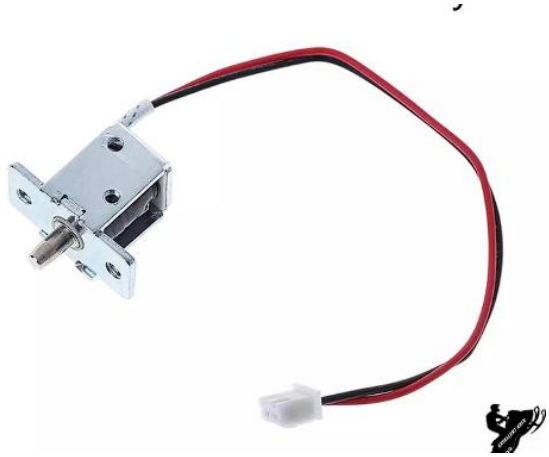
Figure 1: Spiral Model (SDLC)

Using the Spiral Method allows the project to continue to develop ways of solving the technical challenges of combining multiple hardware modules with software logic. Aspects of the project that had the higher risk like making sure that the LM2596 buck converter distributed power reliably or that the UNO and ESP32-CAM talked to each other reliably were able to be developed and tested earlier. The Spiral Method is iterative so that each subsystem could be built and worked on efficiently prior to the final integration, resulting in a reliable and secure door access system.

3.3 Hardware and Software Requirements

Hardware

 Figure 2: Arduino UNO R3	• Serves as the main microcontroller for controlling the RFID module, relay, buzzer, and communication with the ESP32-CAM. Executes programmed logic for door access control.
 Figure 3:ESP32-CAM (with OV2640 camera)	• Captures images or videos of individuals accessing the door, enabling visual verification. Also supports Wi-Fi for image transfer or remote monitoring.
 Figure 4: MFRC-522 RFID Module	• Reads RFID cards and key fobs to identify authorized users for door access. Communicates with Arduino UNO via SPI interface.

 Figure 5: 1-Channel 5V Relay Module	• Acts as an electrical switch controlled by the Arduino UNO to turn the electromagnetic lock on or off, ensuring secure access control.
 Figure 6: DC 12V Electromagnetic Lock	• Secures the door and unlocks when the relay is activated. Operates using electromagnetic force to hold the door closed until access is granted.
 Figure 7: LM2596 Buck Converter	• Steps down voltage from 12V power supply to the required voltage levels (e.g., 5V) for components such as Arduino UNO, ESP32-CAM, and RFID module.

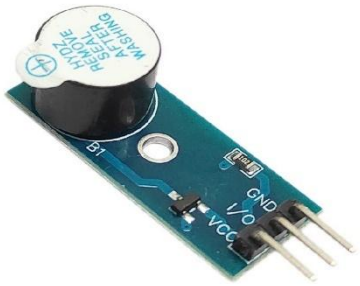
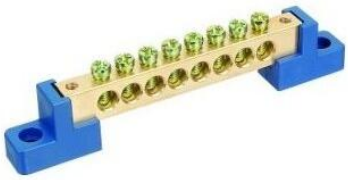
 Figure 8: Buzzer Module	• Provides audio feedback for actions such as successful access, access denial, or system alerts.
 Figure 9: Neutral Link 30A	• Serves as a central distribution point for connecting multiple power lines safely in the system.
 Figure 10: DC Power Connectors (Male/Female, Screw Type)	• Allows secure and detachable connections between power supply and various components.

 Figure 11: 12V Power Adapter	• Provides the main DC power source for the system, powering the electromagnetic lock and other components through the buck converter.
 Figure 12: Jumper Wires (Male-Male, Male-Female, Female-Female)	• Used for making flexible signal and power connections between modules without soldering.
 Figure 13: MBTI Board	• In order to enter the code into ESP32-CAM. Using Micro USB to connect the pc

Software

Arduino IDE

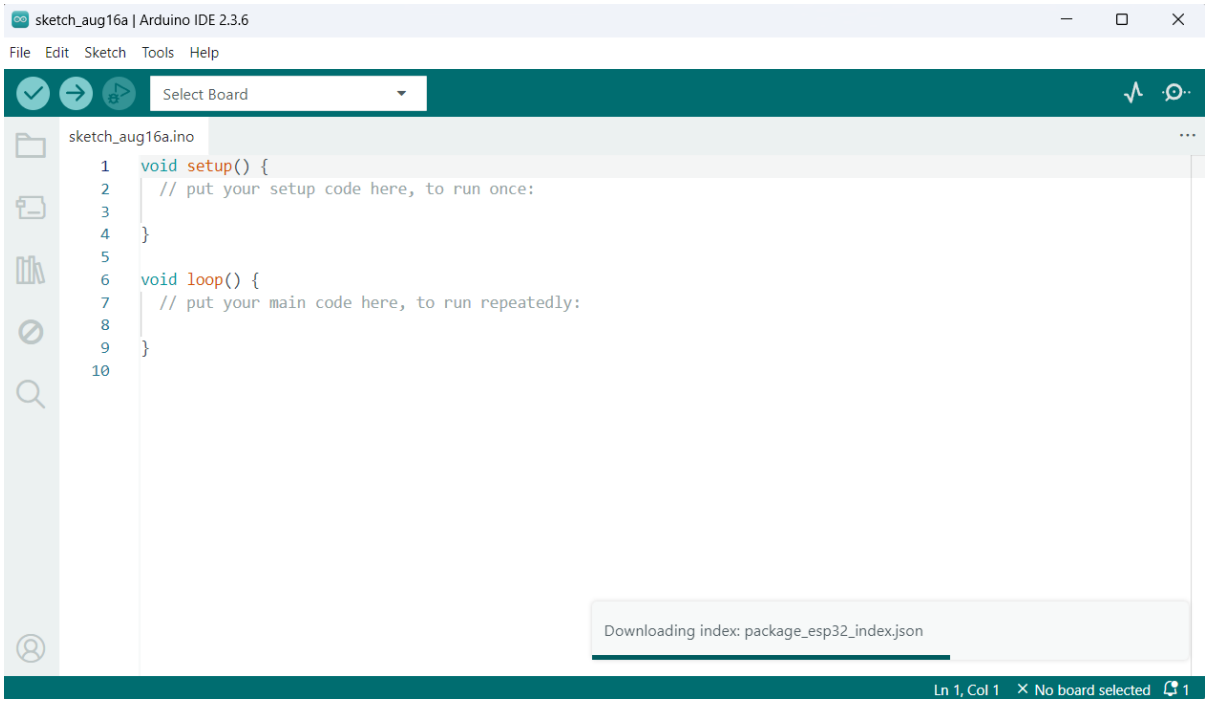


Figure 14: Arduino IDE

- Used to program and upload code into the Arduino UNO and ESP32-CAM.
- Supports C/C++ programming and provides libraries such as MFRC522.h for RFID and esp_camera.h for the ESP32-CAM.

C/C++ Programming Language

- Primary language for developing logic in Arduino UNO and ESP32-CAM.
- Used to implement authentication, camera trigger, relay control, and alarm activation.

Operating System (Linux & Windows)

- Linux is used for coding and testing in Arduino IDE.
- Windows is used for compatibility testing and report documentation.

3.4 Conclusion

In this chapter the methodology, hardware and software requirement were defined and explained, also how the chosen methodology helps with this project. Therefore, in the next chapter we will speak about implementation of this project.

Chapter 4: Requirement Analysis

4.1 Introduction

This chapter contains a comprehensive requirements analysis of the Smart Anti-Theft RFID Door Access System. The primary purpose of requirements analysis is to understand user and system requirements, and to identify and fully specify the features and services that a system will offer. Requirements analysis is critical activity serves to capture requirements and organize information about the system being proposed. One of the steps in the development phase consists of determining the user requirements, which describes a feature of the system, or describes the services the customer should expect from the system. Requirements will be separated between functional and non-functional requirements and accurately describes them; Missteps in this phase can be extremely expensive to change later.

4.2 Gathering and Analysis of Requirements

The wireframes were created based on the findings of observation and analysis of security system requirements for an office or residential location. The participants included potential users (homeowners, staff, etc.) and the system administrator who will be operating the security system. It was noted that several fundamental requirements developed out of the interview, including secure door access through RFID authentication, automated photos taken during access, logging of entry attempts, and emergency alarm engagement. Additional functionality was followed out of the administrator's requirements, including the ability to register or remove RFID tags and maintain a system log.

4.3 Functional Requirement

Requirements	Reasons
RFID Authentication	• Allow users to unlock the door using an authorized RFID card. • Ensure only registered users can gain access.
Camera Capture (ESP32-CAM)	• Capture and store an image whenever an RFID card is scanned. • Provide visual evidence for both successful and failed access attempts.
Relay Control	• Activate or deactivate the electromagnetic lock based on authentication results.
Admin Registration/Removal of RFID Cards	• Allow administrator to add or remove authorized RFID cards in the system.
Access Log Management	• Record user ID, timestamp, and access result (success/failed). • Store logs in SD card or database for monitoring.
Power Management	• Ensure stable operation of all modules through LM2596 power regulation. Allow admin to login to control games.

Table 2: Functional Requirement

4.4 Non-Functional Requirement

Requirements	Reasons
Security	• Ensure only valid RFID tags can unlock the door. • Protect data stored in logs from unauthorized access.
Reliability	• System must operate continuously without frequent crashes or malfunctions.
Performance	• RFID scanning and door unlocking should respond within 2–3 seconds.
Maintenance	• Admin should be able to update RFID card lists and clear logs without interrupting system operation.
Usability	• System should be user-friendly with simple card-tapping operation.
Scalability	• System should allow adding more users without major hardware/software changes.

Table 3: Non-Functional Requirement

4.5 Conclusion

In this chapter requirements were analyzed and acquired through the interview with the admin of a website. Requirements were identified as functional requirements and non-functional requirements. And in the next chapter (Design) the design of the system well be made.

Chapter 5: Design

5.1 Introduction

In system design, the requirements of the Smart Anti-Theft Door Access System are represented in a clear structure with a definition on how the individual hardware and software components (i.e. RFID module, ESP32-CAM, relay, lock, etc) are working together. The design describes system architecture, data flow, and interfaces which provides an overview for implementation and testing.

5.2 System Flow Chat

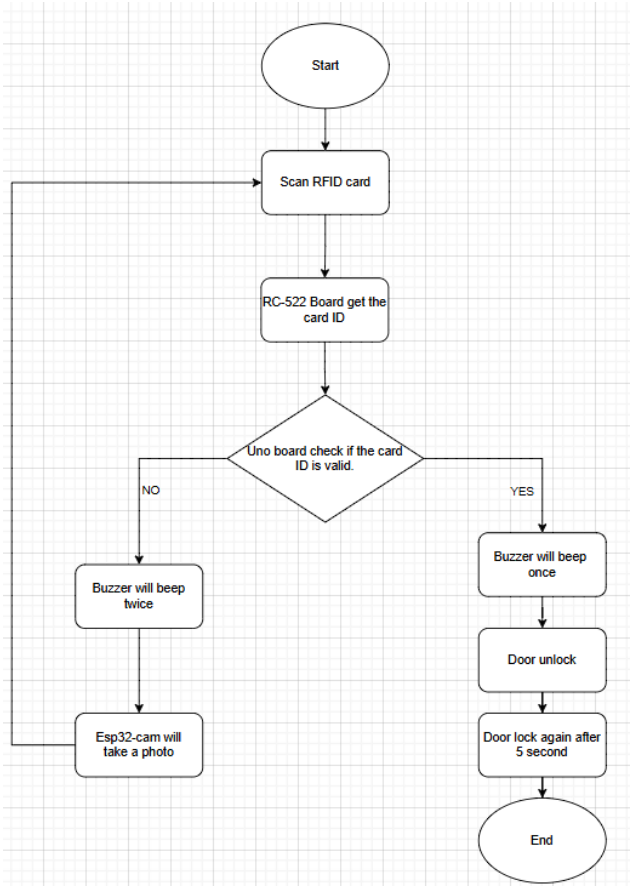


Figure 15: System Flow Chat

The system flowchart provides an overview of the Smart Anti-Theft Door Access System operation. The process starts with a user scanning an RFID card at the RC522 reader. The RC522 module then reads the individual card ID and transmits it to the Arduino Uno for authentication. Now that the Uno has received the information from the RC522 module, it will determine whether the scanned card ID is validated in the system. If the card is authenticated as valid, the buzzer will beep once to indicate the user has successfully authenticated. The

electromagnetic lock in the door will now be released, allowing the door to unlock. The lock will re-engage automatically after a 5-second delay for security, completing the program. In the case that the card outcome is invalid the buzzer will beep twice indicating that there was an error. At the same time the ESP32-CAM module will be triggered to document and take a photo of the person attempting access. This will improve security by recording unauthorized access attempts. After an invalid access attempt, the system then resets and waits for the next RFID scan. This flow allows for security and automation utilizing RFID authentication and camera monitoring, alarm signaling and automatic door relocking.

5.3 Wiring Diagram

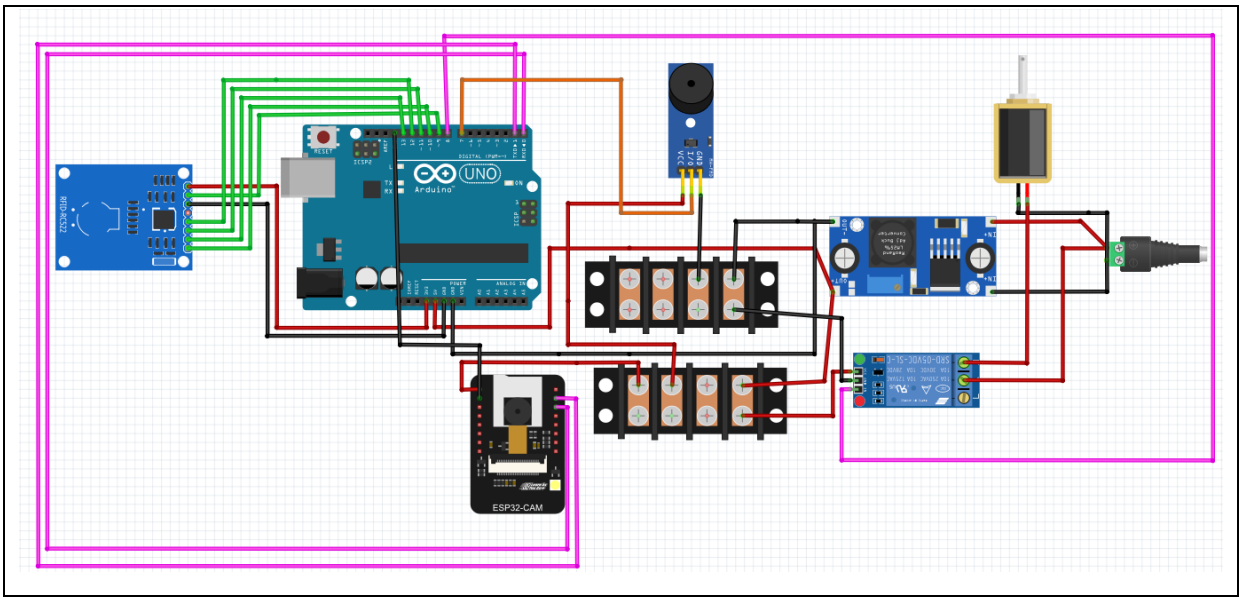


Figure 16: Wiring Diagram

Figure 16 shows the wiring diagram shows all the hardware of the system.

5.4 Signal Wiring Table

UNO	
D9	RC-522: RST
D10	RC-522: SDA
D11	RC-522: MOSI
D12	RC-522: MISO
D13	RC-522: SCK
D7	ALARM: I/O
D8	RELAY: IN1
D0	ESP32-CAM: UOT

D1	ESP32-CAM: UOR

Table 4: Signal Wiring Table

The Arduino UNO acts as the main controller, coordinating input and output signals between different hardware modules. Each pin connection serves a specific function:

- **D9 → RC522 RST (Reset)**
  This pin provides a reset signal to the RC522 RFID reader. It ensures the module can be reinitialized or restarted correctly during operation.
- **D10 → RC522 SDA (Serial Data / Chip Select)**
  Functions as the chip select (CS) line in SPI communication. The UNO uses this pin to enable or disable the RC522 module during data exchange.
- **D11 → RC522 MOSI (Master Out Slave In)**
  Used to send data from the UNO (as SPI master) to the RC522 RFID module (as SPI slave). For example, when the UNO transmits commands or authentication data to the RFID.
- **D12 → RC522 MISO (Master In Slave Out)**
  Used to receive data from the RC522 back to the UNO, such as the unique RFID card ID.
- **D13 → RC522 SCK (Serial Clock)**
  Provides the clock signal required for synchronization in SPI communication between the UNO and the RC522 module.
- **D7 → Alarm (I/O Control)**
  Configured as a digital output pin. When unauthorized access is detected, the UNO sends a HIGH signal to this pin, activating the buzzer alarm module.
- **D8 → Relay IN1**
  This pin controls the relay module. When the UNO outputs a HIGH signal, the relay is activated, switching the state of the electromagnetic lock (lock or unlock).
- **D0 (RX) → ESP32-CAM UOT (TX)**
  Connects the UNO's receive pin to the ESP32-CAM transmit pin. This allows the UNO to receive data or commands from the ESP32-CAM.
- **D1 (TX) → ESP32-CAM UOR (RX)**
  Connects the UNO's transmit pin to the ESP32-CAM receive pin. This allows the UNO to send control signals (e.g., trigger to capture an image) to the ESP32-CAM.

5.5 Power Wiring Table

12v adapter		**TO**	Jack Female Screw terminal
Jack Female Screw terminal	**+/- V**	**TO**	LM2596
	+V	**TO**	Relay COM terminal
	-V	**TO**	Electromagnetic Lock
LM2596	**+/-V**	**TO**	Arduino UNO
		TO	Neutral Link
Arduino UNO	**5V**	**TO**	RC-522
	GND	**TO**	RC-522
Neutral Link	**+V**	**TO**	Buzzer
		TO	ESP32-Cam
		TO	Relay
	-V	**TO**	Buzzer
			ESP32-Cam
			Relay
Relay	**NO PIN**	**TO**	Electromagnetic Lock

Table 5: Power Wiring Table

The system uses a 12V DC adapter as the main power supply. Since different modules operate at different voltage levels, the LM2596 buck converter is used to step down the voltage to 5V for low-voltage devices. The detailed power distribution is as follows:

12V Adapter → Jack Female Screw Terminal

- The 12V DC input is connected to a screw terminal for easy distribution to different modules.

Jack Female Screw Terminal → LM2596 Input (+/- V)

- The LM2596 receives the 12V input and regulates it down to 5V for the Arduino UNO and other 5V devices.

Jack Female Screw Terminal +V → Relay COM Terminal

- The relay common pin is supplied with 12V. When the relay switches, it controls the 12V power line to the electromagnetic lock.

Jack Female Screw Terminal –V → Electromagnetic Lock

- Provides the ground return path for the lock's circuit.

LM2596 Output (+/- V) → Arduino UNO

- The Arduino UNO is powered by the 5V regulated output from the LM2596.
- This ensures the UNO and its peripherals receive a stable power supply.

LM2596 Output (+/- V) → Neutral Link

- The neutral link acts as a distribution point for multiple 5V devices.

Arduino UNO 5V & GND → RC522 RFID Module

- The RC522 operates at 3.3V/5V, powered directly from the UNO.

Neutral Link +V → Buzzer, ESP32-CAM, Relay

- Distributes regulated 5V supply to these modules.

Neutral Link –V → Buzzer, ESP32-CAM, Relay

- Provides ground connections for these modules.

Relay NO Pin → Electromagnetic Lock

- The electromagnetic lock is connected to the relay via the **Normally Open (NO)** terminal.
- In the normal state (no signal), the lock remains unpowered (locked).
- When the relay is triggered, the NO contact closes, supplying 12V to the lock, thus unlocking it.

5.4 ESP32-CAM Code Design

sketch_aug17a.ino

```
#include "esp_camera.h"
#include "FS.h"
#include "SD_MMC.h"

#define RX_PIN 3  // ESP32-CAM默认串口RX
#define TX_PIN 1  // ESP32-CAM默认串口TX

// 摄像头配置，AI Thinker模块引脚定义
#define CAMERA_MODEL_AI_THINKER
#include "camera_pins.h"

void setup() {
  Serial.begin(115200);
  Serial.println("ESP32-CAM start");

  // 摄像头初始化
  camera_config_t config;
  config.ledc_channel = LEDC_CHANNEL_0;
  config.ledc_timer = LEDC_TIMER_0;
  config.pin_d0 = Y2_GPIO_NUM;
  config.pin_d1 = Y3_GPIO_NUM;
  config.pin_d2 = Y4_GPIO_NUM;
  config.pin_d3 = Y5_GPIO_NUM;
  config.pin_d4 = Y6_GPIO_NUM;
  config.pin_d5 = Y7_GPIO_NUM;
  config.pin_d6 = Y8_GPIO_NUM;
  config.pin_d7 = Y9_GPIO_NUM;
  config.pin_xclk = XCLK_GPIO_NUM;
  config.pin_pclk = PCLK_GPIO_NUM;
  config.pin_vsync = VSYNC_GPIO_NUM;
  config.pin_href = HREF_GPIO_NUM;
  config.pin_sscb_sda = SIOD_GPIO_NUM;
  config.pin_sscb_scl = SIOC_GPIO_NUM;
  config.pin_pwdn = PWDN_GPIO_NUM;ZS
  config.pin_reset = RESET_GPIO_NUM;
```

Figure 17: ESP32-CAM Code 1

```
  config.pin_reset = RESET_GPIO_NUM;
  config.xclk_freq_hz = 20000000;
  config.pixel_format = PIXFORMAT_JPEG;
  config.frame_size = FRAMESIZE_VGA;
  config.jpeg_quality = 12;
  config.fb_count = 1;

  esp_err_t err = esp_camera_init(&config);
  if (err != ESP_OK) {
    Serial.printf("cam failed: 0x%x\n", err);
    while (true) delay(1000);
  }
  Serial.println("cam good");

  // 初始化SD卡
  if(!SD_MMC.begin()){
    Serial.println("SD card fail");
    while(true) delay(1000);
  }
  Serial.println("SD card good");
}

void loop() {
  if (Serial.available()) {
    char cmd = Serial.read();
    if (cmd == 'T') {
      Serial.println("photoing");
      camera_fb_t *fb = esp_camera_fb_get();
      if (!fb) {
        Serial.println("photo fail");
        return;
      }

      // 创建一个唯一文件名，格式：photo_时间戳.jpg
      String path = "/photo_" + String(millis()) + ".jpg";
```

Figure 18: ESP32-CAM Code 2

```

      Serial.printf("保存文件：%s\n", path.c_str());

      // 打开文件准备写入
      fs::FS &fs = SD_MMC;
      File file = fs.open(path.c_str(), FILE_WRITE);
      if(!file){
        Serial.println("open file fail");
      } else {
        file.write(fb->buf, fb->len); // 写入图片数据
        Serial.println("save good");
      }
      file.close();

      esp_camera_fb_return(fb);
    }
  }
}
```

Figure 19: ESP32-CAM Code 3

This program is designed to control the ESP32-CAM module, capture photos upon receiving a serial command, and save them to the onboard SD card. The main components of the code are as follows:

Library Import

- esp_camera.h is used to configure and initialize the ESP32-CAM camera module.
- FS.h and SD_MMC.h are used for handling file operations and accessing the SD card storage.

UART Configuration

- The ESP32-CAM uses its default UART pins, where RX_PIN = 3 and TX_PIN = 1. This allows communication with other devices, such as Arduino UNO, through serial transmission.

Camera Configuration

- The code defines the camera model as AI Thinker and loads the specific pin configuration from camera_pins.h.
- A camera_config_t structure is created to set up various parameters including data pins, clock, synchronization signals, and control pins.
- The camera is set to output images in **JPEG format** with a **resolution of VGA (640x480 pixels)**, medium image quality, and one frame buffer.

Camera Initialization

- The function esp_camera_init() initializes the camera with the provided configuration.
- If initialization fails, the system prints an error message and stops execution.

SD Card Initialization

- The code initializes the SD card using SD_MMC.begin().
- If the SD card cannot be accessed, an error message is shown, and the system halts.

Main Loop Operation

- The program continuously listens for serial input.
- When the command **T** is received, the ESP32-CAM captures a photo.
- The image is saved on the SD card with a unique filename based on the system's timestamp (e.g., photo_12345.jpg).
- The image buffer is then released back to the system to free memory.

Outputs and Feedback

- Throughout the process, serial messages are printed (e.g., *"photoing"*, *"save good"*, or error messages) to provide status updates for debugging and monitoring.

5.5 Arduino UNO Code Design

sketch_aug17a.ino

```
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 10
#define RST_PIN 9
#define BUZZER_PIN 7
#define RELAY_PIN 8

MFRC522 mfrc522(SS_PIN, RST_PIN);

// 授权卡 UID
byte authorizedUID[4] = {0xCE, 0xF7, 0x31, 0x06};

void setup() {
  Serial.begin(115200);  // 硬件串口，波特率115200，和ESP32-CAM一致
  SPI.begin();
  mfrc522.PCD_Init();
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(RELAY_PIN, OUTPUT);

  digitalWrite(BUZZER_PIN, HIGH);
  digitalWrite(RELAY_PIN, HIGH);
  Serial.println("系统启动，等待刷卡...");
}

void loop() {
  if (!mfrc522.PICC_IsNewCardPresent() || !mfrc522.PICC_ReadCardSerial()) {
    return;
  }

  Serial.print("检测到卡片 UID: ");
  for (byte i = 0; i < mfrc522.uid.size; i++) {
    Serial.print(mfrc522.uid.uidByte[i], HEX);
    Serial.print(" ");
  }
```

Figure 20: Arduino UNO Code 1

```
  Serial.println();

  if (isAuthorized(mfrc522.uid.uidByte)) {
    beep(1);
    Serial.println("✅ 授权卡，解锁中...");
    digitalWrite(RELAY_PIN, LOW);
    delay(5000);
    digitalWrite(RELAY_PIN, HIGH);
    Serial.println("🔒 自动上锁完成");
  } else {
    beep(2);
    Serial.println("❌ 未授权卡，发送拍照指令...");
    Serial.println("T");  // 发送字符T给ESP32-CAM
  }

  mfrc522.PICC_HaltA();
  mfrc522.PCD_StopCrypto1();

  waitForCardRemoval();
  mfrc522.PCD_Init();
  Serial.println("✅ 卡片已移开，准备下一次刷卡");
}

bool isAuthorized(byte *uid) {
  for (byte i = 0; i < 4; i++) {
    if (uid[i] != authorizedUID[i]) {
      return false;
    }
  }
  return true;
}

void beep(int times) {
  for (int i = 0; i < times; i++) {
    digitalWrite(BUZZER_PIN, LOW);
```

Figure 21: Arduino UNO Code 2

```
void beep(int times) {
  for (int i = 0; i < times; i++) {
    digitalWrite(BUZZER_PIN, LOW);
    delay(100);
    digitalWrite(BUZZER_PIN, HIGH);
    delay(100);
  }
}

void waitForCardRemoval() {
  while (mfrc522.PICC_IsNewCardPresent() && mfrc522.PICC_ReadCardSerial()) {
    delay(100);
  }
}
```

Figure 22: Arduino UNO Code 3

This program controls the RFID Anti-Theft Door Access System using an Arduino UNO, RC522 RFID reader, relay, buzzer, and communication with the ESP32-CAM.

Library Setup and Pin Definitions

- The program includes the SPI and MFRC522 libraries for RFID communication.
- Pins are defined for the RFID reader (SS_PIN = 10, RST_PIN = 9), the buzzer (BUZZER_PIN = 7), and the relay (RELAY_PIN = 8).

Initialization (setup())

- Serial communication is started at **115200 baud**, matching the ESP32-CAM.
- The RFID reader is initialized using SPI.begin() and mfrc522.PCD_Init().
- The buzzer and relay are set as outputs.
- At startup, both buzzer and relay are set HIGH (default OFF state).

Main Loop (loop())

- The Arduino continuously checks for new RFID cards.
- If a card is detected, its UID (Unique Identifier) is read and printed to the Serial Monitor.

Authorization Check

- The UID of the card is compared with a predefined **authorized card UID** stored in the array authorizedUID.
- If the card matches:
  - The buzzer beeps once.
  - The relay is activated (LOW) for 5 seconds, unlocking the electromagnetic lock.
  - After 5 seconds, the relay is deactivated (HIGH), locking the door again.
- If the card does not match:
  - The buzzer beeps twice.
  - A **command "T"** is sent over the serial connection to the ESP32-CAM, instructing it to take a photo of the unauthorized attempt.

Card Removal Handling

- The system waits until the card is removed before re-initializing the RFID reader for the next scan.
- This prevents repeated scanning of the same card.

Helper Functions

- isAuthorized(): Compares the scanned UID with the authorized UID.
- beep(int times): Makes the buzzer sound one or more times depending on the situation.

- waitForCardRemoval(): Ensures the system only responds when a new card is presented.

5.6 Conclusion

In conclusion, this part described in detail how the system will be beneficial, what technologies are used, and how this system works.

Chapter 6: Result and Conclusion

6.1 Introduction

In this chapter, I will show the final outcome of the RFID Anti-Theft Door Access System. Besides that, I will talk about the future trend of my system and the conclusion.

6.2 Final Outcome



Figure 23: System Final Outcome

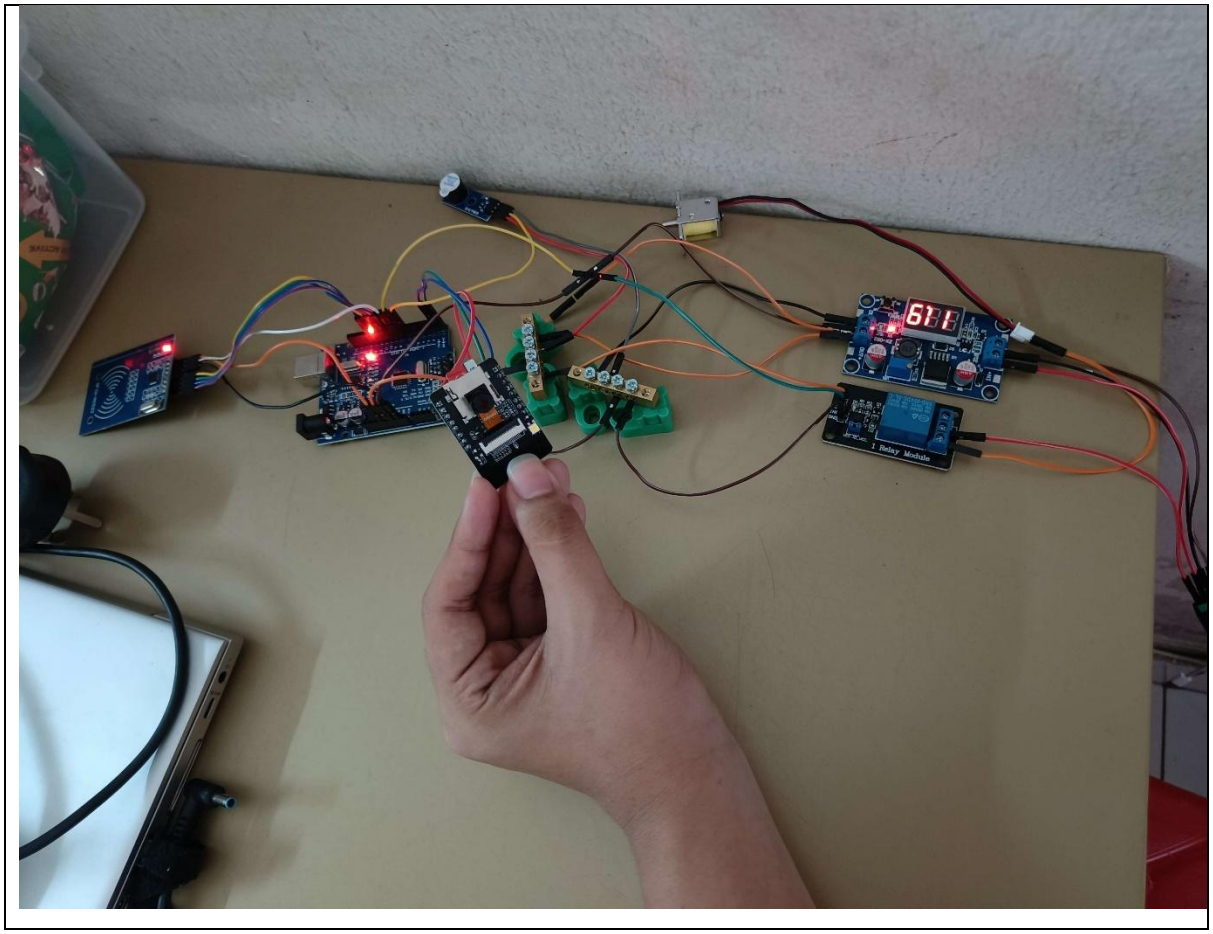


Figure 24: System Final Outcome

6.3 System Future Trends

Add Face Recognition with ESP32-CAM

- Currently, you only use RFID for access.
- You can upgrade to face recognition authentication to prevent card cloning.
- Combine RFID + Face ID for two-factor authentication.

Mobile App Integration

- Create a mobile app (Android/iOS) to control and monitor the system.
- Features:
    - Unlock door remotely.
    - Receive alerts if unauthorized access is attempted.
    - View captured images from ESP32-CAM in real time.

Cloud or Database Connectivity

- Instead of storing access logs only in Arduino/SD card, push them to a cloud database (e.g., Firebase, MySQL).

6.4 Conclusion

This project successfully developed a Smart Anti-Theft Door Access System that combines RFID authentication, camera surveillance, and electronic locking for enhanced security. By integrating Arduino UNO, ESP32-CAM, RC522 RFID, relay, buzzer, and LM2596 power regulation, the system provides a reliable and automated solution for controlling door access. Authorized users can unlock the door through RFID cards, while unauthorized attempts trigger alerts and image capture for evidence. The system demonstrates how embedded hardware and IoT concepts can be applied to create a practical, low-cost, and efficient security solution. Overall, the project not only meets its intended objectives but also lays the foundation for future improvements such as mobile app integration, real-time monitoring, and cloud-based access management.

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